PID MAPS Template (Instrumentation)

The PID MAPS template is designed to control a device that is connected to an analog output.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in two models – advanced and standard, which provide respectively lower levels of control over the PID.

This document describes the Advanced PID MAPS Template and also defines the level of functionality provided by the standard model. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the device being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced PID Functional Philosophy

The PID Modified Value output is enabled once all setpoints are set and the start request is received.

During PID control, the value measured by the sensor (process value) is compared with the preset value (set value). The output value (manipulated value) is then adjusted in order to eliminate the difference between the process value and the set value. The MV (manipulated value) is calculated by combining the proportionate operation (P), the integrating operation (I), and the differentiating operation (D) so that the PV is brought to the same value as the SV quickly and precisely. The MV is made large when the difference between the PV and the SV is large so as to bring the PV close to the SV quickly. As the difference between the PV and the SV gets smaller, a smaller MV is used to bring the PV to the same value as the SV gradually and accurately.

PID Setpoints

- PID Setpoint
- P Setpoint
- I Setpoint
- D Setpoint
- Sample Setpoint
- In Filter Setpoint
- MV Lim Hi
- MV Lim Lo
- Raw Min
- Raw Max
- Eng. Min
- Eng. Max

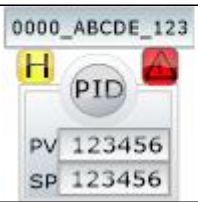
PLC Interlocks

- Start Interlock Ready

Start requests are initiated by selecting a mode on the PID faceplate:

- **Auto mode** - the auto control program set the MV value.
- **Manual mode** - the operator can set a fixed MV value from the faceplate.

Typical Faceplate Graphics for the Advanced PID MAPS Template

PID	
	
PV	PID Process Value
SP	PID Setpoint Value (Preset value)
	PID Control Name
	Manual Mode
	Fault Indication

Signal Description		**Agent Type**	**Advanced**	**Standard**	**Basic**
SCADA Control Words:					
SSCL 0	Control Word	**Marshal**	x	x	
SSCL 1	PID Setpoint	**Analog**	x	x	
SSCL 2	P Setpoint	**Analog**	x	x	
SSCL 3	I Setpoint	**Analog**	x	x	
SSCL 4	D Setpoint	**Analog**	x	x	
SSCL 5	Sample Setpoint	**Analog**	x	x	
SSCL 6	In Filt Setpoint	**Analog**	x	x	
SSCL 7	MV Lim Hi	**Analog**	x	x	
SSCL 8	MV Lim Lo	**Analog**	x	x	
SSCL 9	Mv Man Setpoint	**Analog**	x	x	
SSCL 10	Raw Min	**Analog**	x		
SSCL 11	Raw Max	**Analog**	x		
SSCL 12	Eng Min	**Analog**	x		
SSCL 13	Eng Max	**Analog**	x		

Signal Description		**Agent Type**	**Advanced**	**Standard**	**Basic**
SCADA Status Words:					
SSSL 0	Status Word	**Marshal**	x	x	
SSSL 1	PID Process Value	**Analog**	x	x	
SSSL 2	PID Manipulated Value	**Analog**	x	x	

Analog Input:		**Advanced**	**Standard**	**Basic**
AI 0	PID Process Value	x	x	
Analog Output:				
AO 0	PID Modified Value	x	x	

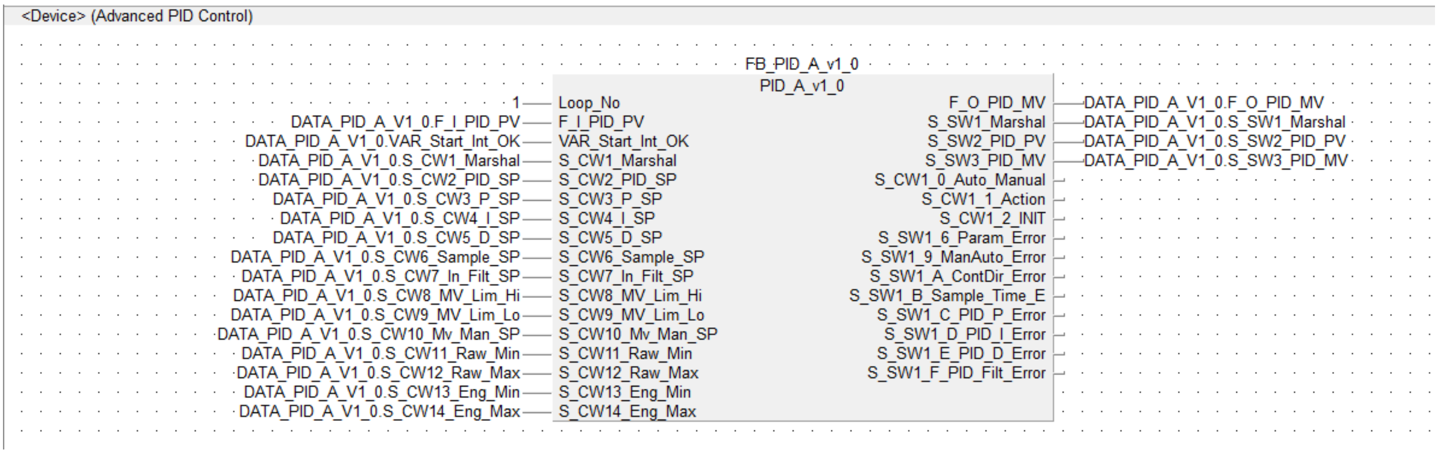
SCADA Control Word (Adroit Marshal Agent)		**Event Logged**	**Advanced**	**Standard**	**Basic**
Bit 0	Auto Manual	x	x	x	
Bit 1	Control Direction	x	x	x	
Bit 2	Initiate	x	x	x	

Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Spare				
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

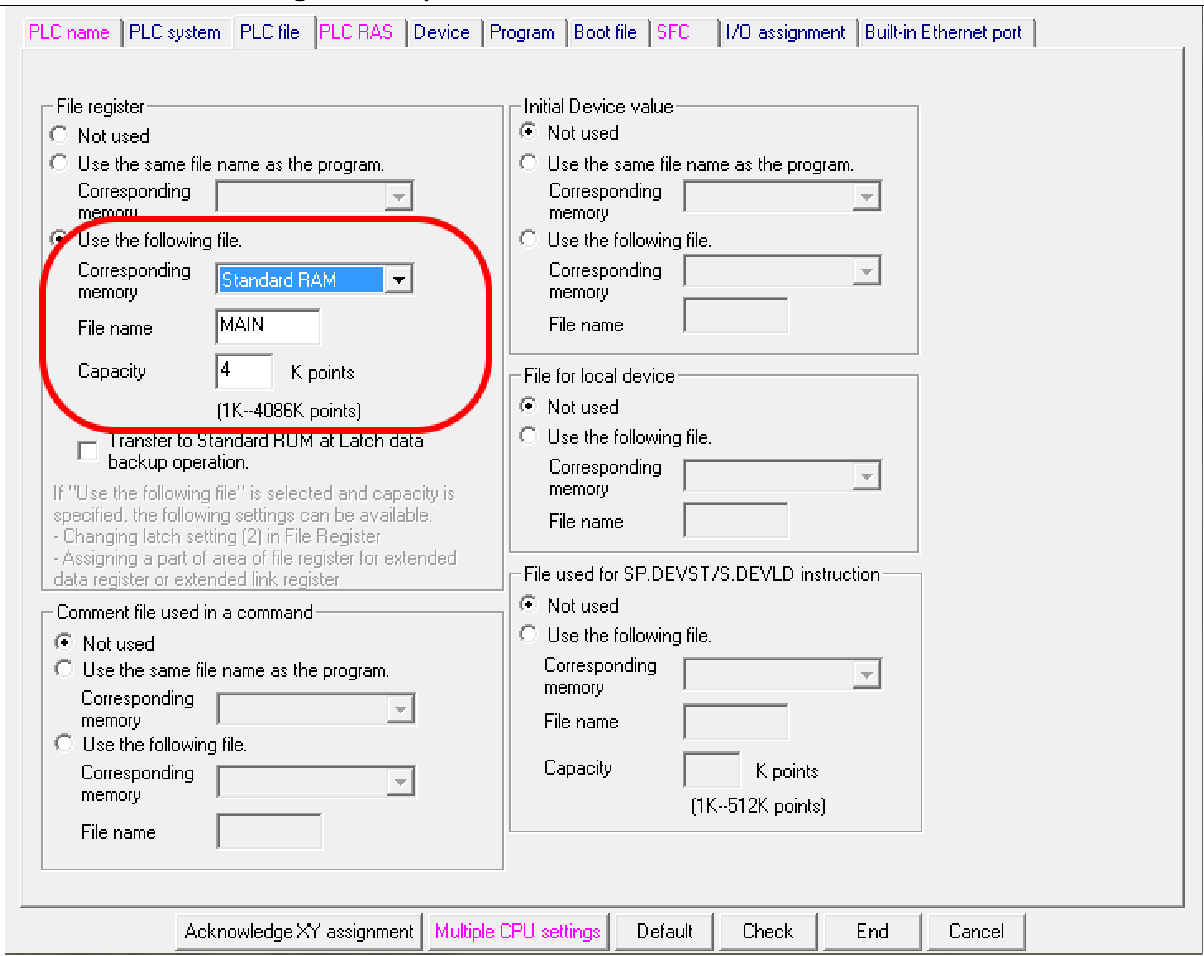
SCADA Status Word (Adroit Marshal Agent)		**Alarmed**	**Advanced**	**Standard**	**Basic**
Bit 0	Spare				
Bit 1	Spare				
Bit 2	Spare				
Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Parameter Fault	**x**	**x**	**x**	
Bit 7	Spare				
Bit 8	Spare				
Bit 9	Auto Man Fault	**x**	**x**	**x**	
Bit 10	Control Direction Fault	**x**	**x**	**x**	
Bit 11	Sample Time Fault	**x**	**x**	**x**	
Bit 12	Proportional Band Fault	**x**	**x**	**x**	
Bit 13	Integral Band Fault	**x**	**x**	**x**	
Bit 14	Derivative Band Fault	**x**	**x**	**x**	
Bit 15	Filter Fault	**x**	**x**	**x**	

Resources	**Advanced**	**Standard**	**Basic**
Digital Inputs	0	0	
Digital Outputs	0	0	
Analog Inputs	1	1	
Analog Outputs	1	1	
PLC Memory Steps	412	250	
System Words Used	46	11	
System Bits Used	74	68	
SCADA SCAN Points	17	13	

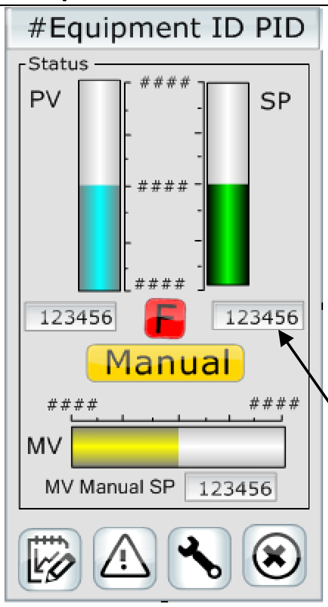
Template: Advanced PID PLC Function Block



Setup: Advanced PID PLC File Register Setup

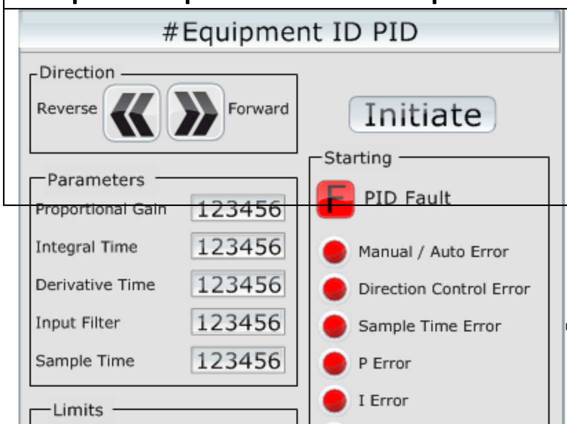


Template Graphic: Advanced Home Screen



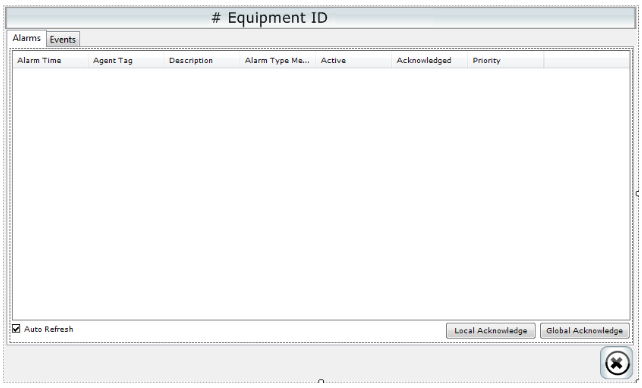
Control	Descriptions
#Equipment ID PID	PID Control Name
Manual	Manual control MV value by operator
MV Manual SP	Manual Setpoint for MV (Manipulated Value)
	Manual control from the field start/stop station
PV	PID Process Value
SP	PID Setpoint Value
PID Setpoint	Sets the PID Control Setpoint Value

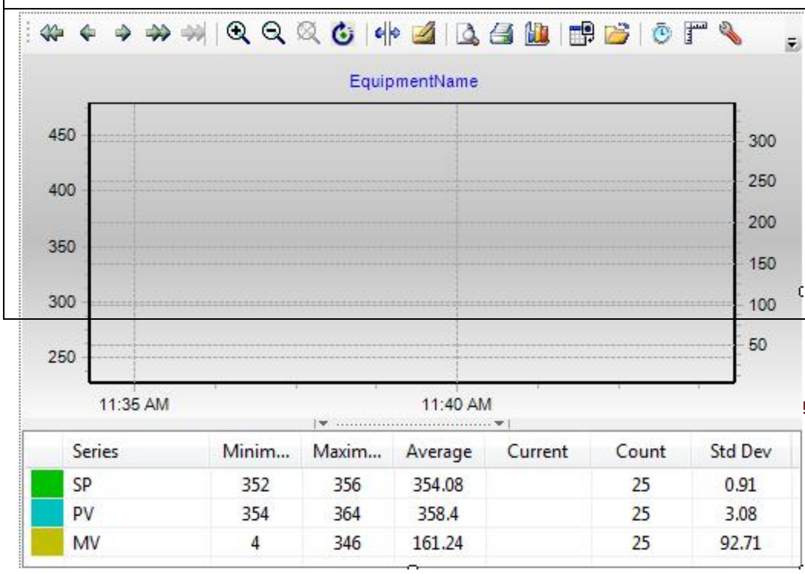
Template Graphic: Advanced Setup Screen



Control	Descriptions
Initiate	Initiates the PID once a Setpoint have changed
Proportional Gain	Proportionate Operation (P) Setpoint
Integral Time	Integrating Operation (I) Setpoint
Derivative Time	Differentiating Operation (D) Setpoint
Sample Time	Sampling Cycle (Ts) Setpoint

	Input Filter	Filter Coefficient Setpoint
	MV High Limit	Manipulated Value High Limit (Mv Lim Hi) Setpoint
	MV Low Limit	Manipulated Value Low Limit (Mv Lim Lo) Setpoint
	Raw. Min	Scale Raw Minimum Input
	Raw. Max	Scale Raw Maximum Input
	Eng. Min	Scale Engineering Minimum Output
	Eng. Max	Scale Engineering Maximum Output
	Forward	Forward operation, the MV (manipulated value) increases as the PV (process value) increases beyond the SV (set value).
	Reverse	Reverse operation, the MV (manipulated value) increases as the PV (process value) decreases below the SV (set value).

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Name of controlled PID
	SP	Trend of PID Setpoint Value (Preset value)

	PV	Trend of PID Process Value (Input value)
	MV	Trend of PID Manipulated Value (Output value)